

TIC 41996755

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS

Period 9.563067 d
Epoch (t0) 1683.9149 BTJD
Duration 0.1200 d
Depth 10.895 % (108,950 ppm)
SNR 290.0

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

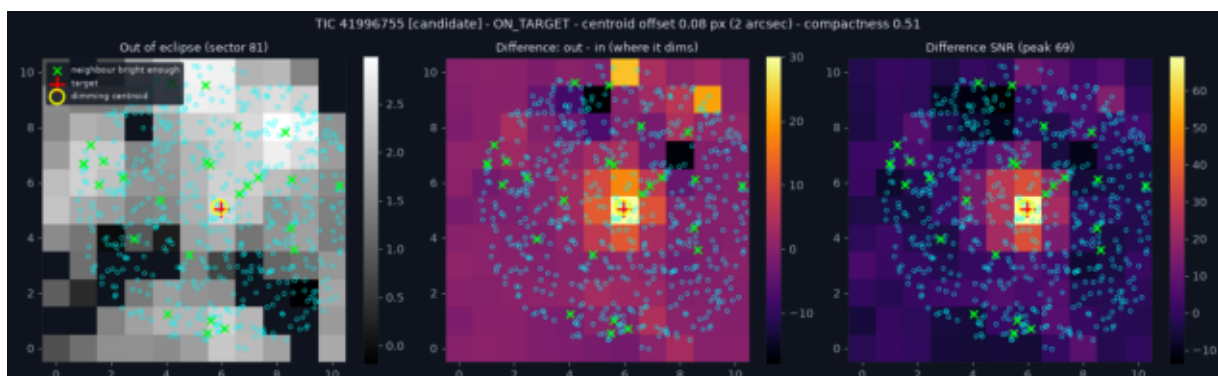
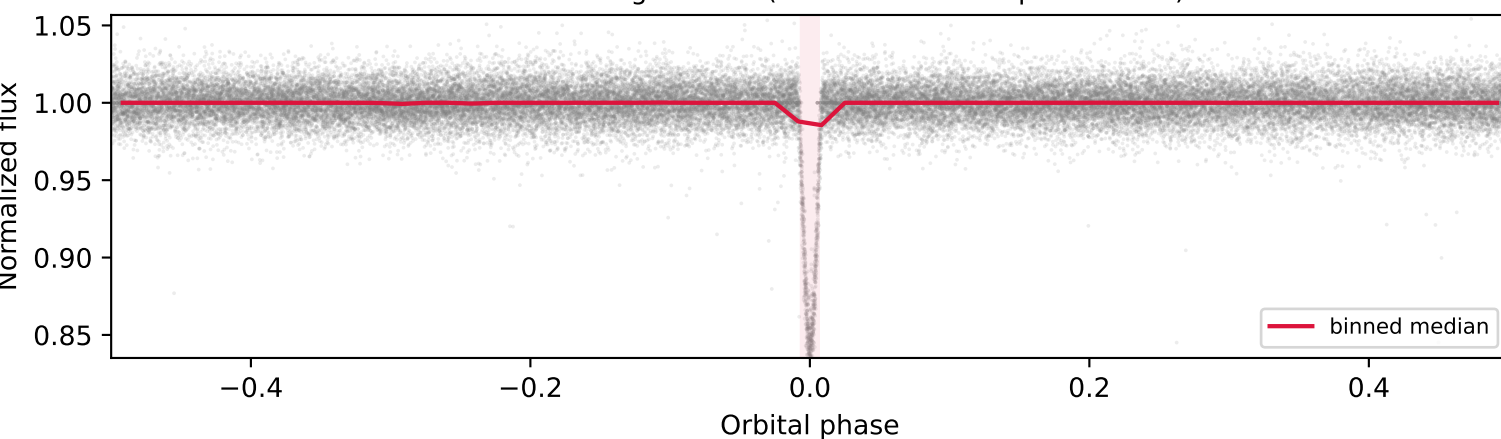
Eclipses observed 24 (gate ≥ 3)
Robust depth ratio 1.11 (gate ≥ 0.50)
Robust depth sigma 245.0 (gate ≥ 7.0)
Odd/even z-score 0.98 (consistent)
Secondary eclipse not detected

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 81 light-curve dip in this sector: 102.7 sigma t0 refined by 0.0000 phase
Difference-image peak SNR 69.47 (gate ≥ 5.0) Compactness 0.51 (gate ≥ 0.50)
Dimming centroid offset 0.08 px (2 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 41996781
0.84 px away, dTmag +0.52 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)



No caveats - clean detection on every check.